Reinforcement Learning or Transformers for Textual Data

Pick one route: a model that reads text, or an agent that learns from rewards

Text | Route A: transformers | Route B: reinforcement learning | Uses Weeks 7, 9 and 10

The task. A public organisation receives thousands of short written messages and must send each one to the right team, quickly and fairly. In **Route A** you build a model that reads a message and predicts which team it belongs to. In **Route B** you build an agent that learns by trial and error which action to take for each message, guided only by rewards.

THE DATA

In Week 9, the teaching team gives you a collection of recent short messages, each labelled with the team that handled it. They were collected recently, so no AI assistant has seen them. Route B also gets a small simulated environment that gives the agent a reward for each choice it makes. Messages may contain personal details: keep them on your computer, and hide names in any example you show.

WHAT YOU WILL DO

Route A, transformers. Start with a simple word-count model (Week 2). Take a small pretrained transformer and adapt it, first by training only a new final layer, then by fine-tuning the whole model (Week 9). Show where it pays attention, and test what each layer has learned (Week 10).

Route B, reinforcement learning. Build a small network that reads a message and picks an action, and train it from rewards with the REINFORCE method. Compare it with an ordinary classifier, and plot the reward as it learns. This route goes beyond the lectures, so choose it only if you are happy to read ahead (Sutton and Barto, Chapter 13).

WHAT YOU CODE YOURSELF

Route A: backpropagation and an optimiser step for the final layer you add to the transformer. **Route B:** backpropagation of the REINFORCE loss and the optimiser step. **Both routes:** PCA via SVD to map the messages in two dimensions. Test each piece against a library.

IS IT FAIR AND SAFE?

Check whether the model works equally well for different groups of writers, for example people writing in different varieties of English. Try one fix and measure what it costs. Use LIME or SHAP to show which words drove a decision. In Route B, also ask whether the reward itself is fair to everyone. Write a note on personal data in messages, and on who checks the model's mistakes.

QUESTIONS TO PREPARE FOR

Route A: what is one attention head looking at, and does that explain the decision? Route B: why does the agent need so many tries, and what is your reward really encouraging? Both: why not just use a general chatbot?

COMMON MISTAKES

Comparing a carefully tuned transformer with a baseline you did not tune. The same message appearing in both training and test data. In Route B, a reward the agent can earn without doing the task.

WEEK-BY-WEEK PLAN

Week 9: simple baseline trained; route chosen; for Route B, the environment working with random actions. **Week 10:** transformer adapted, or agent learning; hand-written pieces passing their tests. **Week 11:** attention and layer tests, or reward curves against simple strategies; fairness check with one fix; model card. **By 6 January 2027:** report written; repository checked and submitted. **Week of 11 January:** 10-minute in-person interview.

TO AIM HIGHER

Route A: compare LoRA, a cheap way to fine-tune, with full fine-tuning on accuracy, size and time, and test the model on messages full of typos. Route B: subtract a running average from the reward to steady the learning, and compare with random and greedy strategies. Both: study the twenty messages the model gets most confidently wrong.